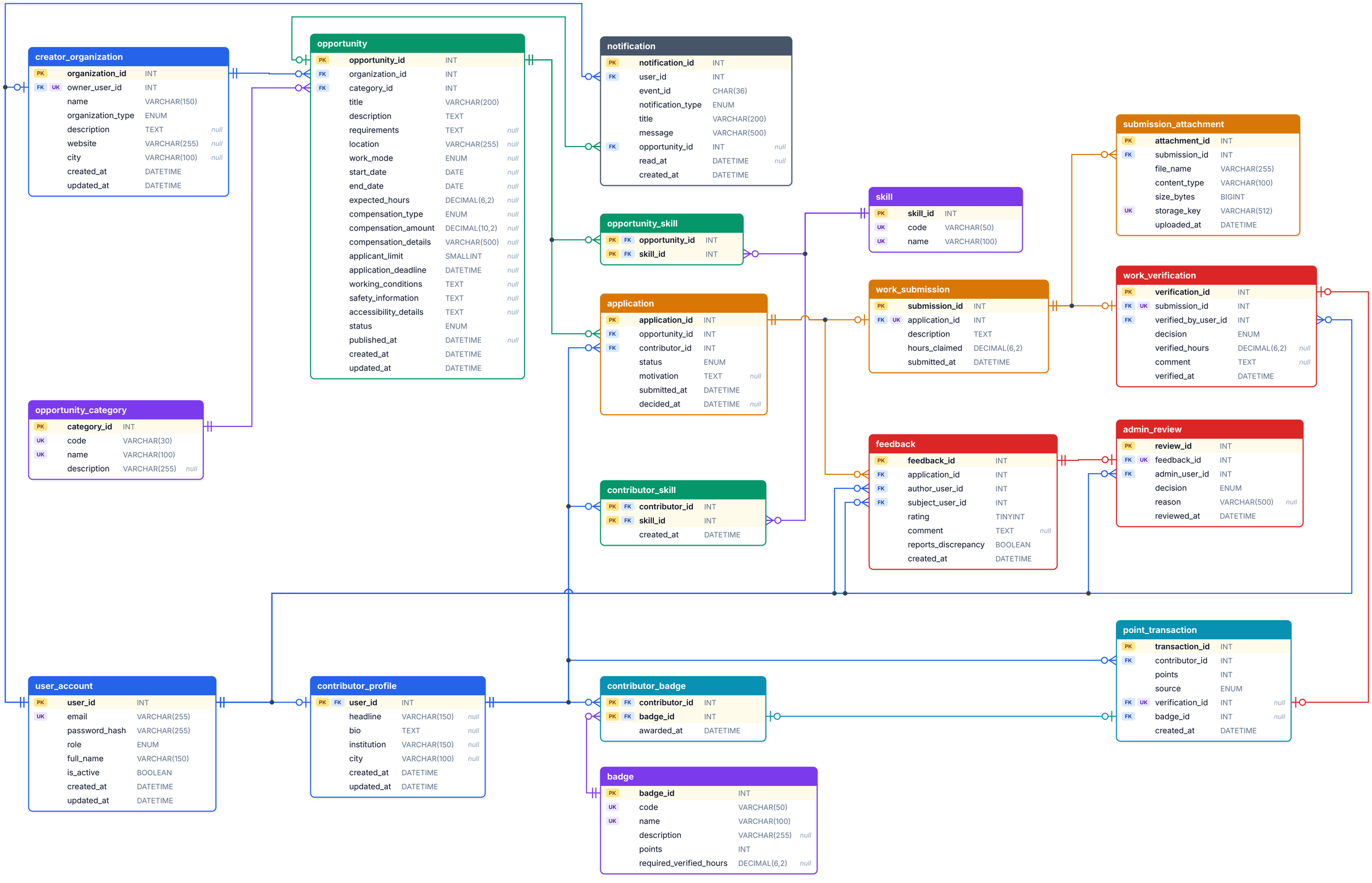
GROW8 — Entity-Relationship Diagram

MySQL 8.4 · 18 tables · 124 columns · 26 foreign keys · 17 integrity triggers · 5 derived-value views · all tables in BCNF

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Cardinality (crow's foot)

- **Exactly one** — parent side, FK is NOT NULL
- **Zero or one** — parent side if FK is nullable; child side if FK is UNIQUE
- **Zero or many** — child side of an ordinary FK
- **Hop** — lines cross, not connected
- **Branch** — one key, several tables

Columns

- PK — Primary key
- FK — Foreign key
- UK — Unique key
- *null* — Nullable (every other column is NOT NULL)

INT = INT UNSIGNED · DATETIME in UTC · BOOLEAN = TINYINT(1)
Lines run from the parent key to the exact FK column.
point_transaction → contributor_badge is the composite key (contributor_id, badge_id).

Table groups (line colour = parent group)

- Identity and profiles
- Reference data
- Opportunities and skills
- Applications and work
- Verification and feedback
- Rewards
- Notifications

Derived values: views, never stored

v_contributor_skill
verified once approved work used the skill

v_contributor_verified_hours
SUM(verified_hours) of approved work_verification rows

v_feedback_status
pending until an admin_review row exists

v_leaderboard
SUM(points) from point_transaction, RANK() OVER

v_user_rating
AVG(rating) of feedback approved in admin_review

GROW8 — Relational Schema

Primary keys underlined · foreign keys in blue italics, each listed under its relation as columns → relation(key) · all relations in BCNF

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IDENTITY AND PROFILES

user_account(**user_id**, email, password_hash, role, full_name, is_active, created_at, updated_at)

contributor_profile(***user_id***, headline, bio, institution, city, created_at, updated_at)
- user_id → user_account(user_id)

creator_organization(**organization_id**, *owner_user_id*, name, organization_type, description, website, city, created_at, updated_at)
- owner_user_id → user_account(user_id)

REFERENCE DATA

opportunity_category(**category_id**, code, name, description)

skill(**skill_id**, code, name)

badge(**badge_id**, code, name, description, points, required_verified_hours)

OPPORTUNITIES AND SKILLS

opportunity(**opportunity_id**, *organization_id*, *category_id*, title, description, requirements, location, work_mode, start_date, end_date, expected_hours, compensation_type, compensation_amount, compensation_details, applicant_limit, application_deadline, working_conditions, safety_information, accessibility_details, status, published_at, created_at, updated_at)
- category_id → opportunity_category(category_id)
- organization_id → creator_organization(organization_id)

opportunity_skill(***opportunity_id***, ***skill_id***)
- opportunity_id → opportunity(opportunity_id)
- skill_id → skill(skill_id)

contributor_skill(***contributor_id***, ***skill_id***, created_at)
- contributor_id → contributor_profile(user_id)
- skill_id → skill(skill_id)

APPLICATIONS AND WORK

application(**application_id**, *opportunity_id*, *contributor_id*, status, motivation, submitted_at, decided_at)
- contributor_id → contributor_profile(user_id)
- opportunity_id → opportunity(opportunity_id)

work_submission(**submission_id**, *application_id*, description, hours_claimed, submitted_at)
- application_id → application(application_id)

submission_attachment(**attachment_id**, *submission_id*, file_name, content_type, size_bytes, storage_key, uploaded_at)
- submission_id → work_submission(submission_id)

VERIFICATION AND FEEDBACK

work_verification(**verification_id**, *submission_id*, *verified_by_user_id*, decision, verified_hours, comment, verified_at)
- submission_id → work_submission(submission_id)
- verified_by_user_id → user_account(user_id)

feedback(**feedback_id**, *application_id*, *author_user_id*, *subject_user_id*, rating, comment, reports_discrepancy, created_at)
- application_id → application(application_id)
- author_user_id → user_account(user_id)
- subject_user_id → user_account(user_id)

admin_review(**review_id**, *feedback_id*, *admin_user_id*, decision, reason, reviewed_at)
- admin_user_id → user_account(user_id)
- feedback_id → feedback(feedback_id)

REWARDS

contributor_badge(***contributor_id***, ***badge_id***, awarded_at)
- badge_id → badge(badge_id)
- contributor_id → contributor_profile(user_id)

point_transaction(**transaction_id**, *contributor_id*, points, source, *verification_id*, *badge_id*, created_at)
- contributor_id → contributor_profile(user_id)
- contributor_id, badge_id → contributor_badge(contributor_id, badge_id)
- verification_id → work_verification(verification_id)

NOTIFICATIONS

notification(**notification_id**, *user_id*, event_id, notification_type, title, message, *opportunity_id*, read_at, created_at)
- opportunity_id → opportunity(opportunity_id)
- user_id → user_account(user_id)

VIEWS (DERIVED, NOT STORED)

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